0.1 Verification record and remaining boundary

Cumulative fifth-block evidence

Fresh Lean checks. A separate baseline rebuild accepted the unchanged 23-module, 247-declaration fourth block. The final run rebuilt all 30 active modules into a new output directory under Lean 4.34.0-rc2 and Mathlib revision

`aeb72f1e9981b6d35b426b16dca7759be3e7ba06`.

All modules returned exit code zero with warnings treated as errors. All 313 theorem and 36 construction audit targets matched the inventory. Dependencies are contained in `propext`, `Classical.choice` and `Quot.sound`. Both deliberately invalid controls were rejected. Seventeen existing verifier tests and twelve new release-contract tests passed.

Preservation and corrections. All 23 upstream proof files and the original manuscript assets are unchanged. The source-to-result ledger is updated, including the explicit conventions for empty graphs, reflexive reachability and nonempty condition-number spaces. Two stale fourth-ledger increment fields were corrected: they had retained older "new items" metadata even though its cumulative count was correct. The new regression tests check these fields against this block's actual targets.

Development failures. Seventeen captured fifth-block development checks contain ten failed drafts and seven accepted drafts, one per new module. Failures included declaration scoping, finite-type instances, rewrite direction, parser syntax and warnings. The source snapshots and logs are retained separately. They are not substituted for the successful fresh cumulative run. Every final active source has a matching successful audit.

Independent mirrors. Exact rational tests passed 820 band-address cases, 200 DAG/staged-solve cases, 60 inverse-support cases, 50 Schur-support cases and 72 non-coordinate projector cases. Of the DAG cases, 125 also exhausted all candidate schedules on at most four vertices. Eight cyclic graphs were rejected. The inverse and Schur tests checked 390 and 97 absent-route blocks respectively, and retained a cancellation control.

Separate numerical tests passed 64 complex projector cases and 120 positive-matrix normalization/CG-factor cases. The largest squared-Frobenius identity residual was 1.42×10^{-13}; the largest positive condition-bound violation was 6.66×10^{-15}, within the declared tolerance. These are independent Python mirrors, not Lean execution, new universal proofs or GPU measurements.

Remaining item. Manuscript theorem 5.6, small-core optimal recompression, remains unformalized. Four external inputs remain separately tracked: Eckart–Young–Mirsky, CG convergence, the floating-point product model and simultaneous spectral decomposition. The register's 55/56 therefore must not be read as verification of every assertion or all classical algorithms in the paper.

Reproduction and limits. Run `scripts/verify.py` with the separately supplied portable kit. The current evidence is `verification/fifth_final/SUMMARY.json`. The original CPU/GPU corpus, earlier compiler execution suites and Papers I–VII in full were not rerun. The pinned Mathlib cache was reused, not independently rebuilt or rechecked from source. No manuscript-to-formal semantic-equivalence theorem is claimed.